

Team 24 Project Backlog

“Studious”

Team Members:

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Problem Statement:

Our app Studious aims to streamline studying and solve the issue of procrastination by organizing all your tasks into an easy-to-use interface with features such as a to-do list, calendar, and AI tools to create unique study plans for every class. While many similar apps already exist, such as Quizlet and Notion, most rely on individual planning and motivation, making them ineffective for students who struggle with procrastination. As such, Studious will include AI note taking and studying features that will allow students to focus more on applying the material, while also hosting a competitive environment that incentivizes studying across connected users based on each individuals' progress. This will differentiate Studious from other studying apps by streamlining and socializing the act of studying to keep students motivated.

Background Info:

Problem

Many students struggle with procrastination despite having access to existing study apps, due to their reliance on self-motivation. Studious aims to overcome this challenge by combining AI-driven study assistance with social elements to promote sustained engagement.

Domain and Target Users

Many students increasingly turn to digital study assistants to organize their coursework, support their learning across multiple subjects, and make learning less time-consuming. However, these apps fail to account for the needs of different students in one way or another. Apps with a pre-built study plan lose out on personalization to different subjects and specific users' needs, and apps that advertise personalization require a lot of setup by the user. This leads to students having to juggle multiple apps for different purposes.

Similar Platforms

There are many productivity/EdTech apps like Quizlet, Forest, UpAhead, and DuoLingo. Each of which aims to empower students in different or limited fields. Quizlet offers a wide range of study tools like flashcards and other games. Forest is a productivity timer, with UpAhead being a time management tool with an AI-powered to-do list and calendar. DuoLingo focuses on learning languages with a greater emphasis on competing with other users.

Limitations and Solutions

While existing apps are useful, they share two critical limitations. First, they lack support for students prone to procrastination. Apps like Quizlet require users to manually create flashcards before studying, which is time-consuming and counterproductive. Second, these apps are focused on specific functions. DuoLingo gamifies education but is mainly catered to language learning only. Forest is engaging, but it ultimately is just a study time tracking, complementing other apps. Instead of saving time, these limitations lead to students managing multiple apps and undermines the very efficiency these apps aim to create.

We aim to address these issues by creating a comprehensive app that enhances these existing features with AI. To mitigate procrastination, Studious will use AI to generate and organize study plans (i.e. generated practice questions) as well as social features that incentivise participation, such as a leaderboard. Studious will help users organize a study routine by having a to-do list, calendar, and reminders that can be generated on uploaded syllabi and course materials. These features will make the app more adaptable to the needs of different people so they don't have to juggle several apps and waste time doing so.

Functional Requirements:

User Creation

1. As a user, I would like to be able to register for a Studious account.
2. As a user, I would like to be able to login to my Studious account.
3. As a user, I would like to be able to reset my password in case I forget it.
4. As a user, I would like to enable two-factor authentication for enhanced account security.
(if time allows)
5. As a user, I would like to update my email address and account settings.
6. As a user, I would like to delete my account and associated data.

Time Management (To-Do List and Calendar)

1. As a user, I would like to be able to manually create, edit, and mark tasks completed via a to-do List.
2. As a user, I would like to be reminded through email of my tasks and deadlines, and be able to change notification settings (when and how often).
3. As a user, I would like to filter tasks by course, due date, and priority.
4. As a user, I would like to assign priority levels (high, medium, low) to my tasks.
5. As a user, I would like AI to suggest optimal study times based on my deadlines and priority so that I can manage my time efficiently. (if time allows)
6. As a user, I would like to export my calendar to integrate with other calendar apps (Google Calendar, Outlook) so that I can sync my schedule. (if time allows)
7. As a user, I would like to manually add, edit, or remove events to my calendar.
8. As a user, I would like to set recurring tasks for regular assignments or study sessions.
9. As a user, I would like to add subtasks to break down larger assignments.
10. As a user, I would like to attach files or links to specific tasks for quick reference.
11. As a user, I would like to set credit hours or workload estimates for each course.

12. As a user, I would like to receive AI suggestions for task breakdown based on assignment complexity.

Study Tools

1. As a user, I would like AI to generate practice questions based on my notes.
2. As a user, I would like to generate flashcards based on study plans with AI.
3. As a user, I would like to manually create and edit the content of flashcards as needed.
4. As a user, I would like AI to generate/update practice tests based on the mastery of previously generated practice questions and flashcards. (if time allows)

Notes

1. As a user, I would like to upload lecture videos to generate AI notes.
2. As a user, I would like to upload lecture slides to generate AI notes.
3. As a user, I would like to generate notes from a lecture in real time using speech-to-text AI. (if time allows)
4. As a user, I would like to use voice input to ask questions to AI or write notes. (if time allows)
5. As a user, I would like to organize notes based on courses and topics.
6. As a user, I would like to use text formatting (font, font size, color, spacing, etc) to make notes easier to review.
7. As a user, I would like to tag notes with keywords to be able to search across notes.

Milestones and Progress Tracking

1. As a user, I would like to see visual progress trackers (e.g, progress bars).
2. As a user, I would like to earn badges or achievements for completing milestones.
3. As a user, I would like to set custom milestones for each course so that I can track specific goals. (if time allows)
4. As a user, I would like to view detailed analytics of my study time by course and week.
5. As a user, I would like to receive weekly summary reports of my productivity.

Social

1. As a user, I would like to be able to customize my public profile (profile picture, username, etc).
2. As a user, I would like to view the progress of other users using the application through a global leaderboard.
3. As a user, I would like to be able to add friends. (if time allows)
4. As a user, I would like to be able to manage my friends. (if time allows)
5. As a user, I would like to set my profile visibility (public, friends-only, private). (if time allows)
6. As a user, I would like to view my friends' progress via their own public profiles. (if time allows)
7. As a user, I would like to individually message my friends or other users. (if time allows)
8. As a user, I would like to create or join study groups within the app. (if time allows)
9. As a user, I would like to share notes with other users or study groups. (if time allows)

User Interface

1. As a user, I would like to upload syllabi or course schedules to generate a study plan using AI.
2. As a user, I would like to view my study plans within their own dedicated interfaces that have course-specific to-do lists and calendars.

Non-Functional Requirements:

Performance

The system should respond to standard user actions (e.g., viewing notes, updating to-do list) within 500 ms under normal load. AI generated lecture summaries and task breakdowns should be produced within 5-10 seconds. Audio transcription should begin processing after 2 seconds of audio upload.

Scalability

The backend should support 100 concurrently without significant performance reduction. The database schema should scale horizontally to support the growth of the user base as well as each user's notes, progress, activities, and social network.

Usability

The interface should be responsive and usable for both mobile and desktop use. A user should be able to create a note, task, or reminder within a few minutes of logging in. The app should provide clear visual feedback for each click on a button/page. The user interface should be well organized, with study plans clearly separated and all components within each plan easily navigable and intuitive to use.

Security

All user authentication should have secure password hashing and JWT-based authentication. Users can only access their own content, notes, and tasks unless explicitly shared. The use of Mongoose to interact with MongoDB prevents the risk of NoSQL injection. Input validation and sanitization shall be applied to all user-provided data to further reduce the risk of malicious input. Using React will protect the system against cross-site scripting by properly encoding user-generated content before rendering it.